Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/27/2005
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Project Title:	Extensional Rheology Experiment (ERE)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4307	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	12/01/1998	**End Date:**	02/08/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	4
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hall, Nancy Rabel	**Contact Phone:**	216-433-5643
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Flight Program:			
Flight Assignment:	note: extended one year		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 3 610		
Performance Goal No.:			
Performance Goal Text:			
	This research effort has involved an initial ground-based experimental design, and is proceeding ultimately to payload implementation as an International Space Station (ISS) Glovebox Investigation, of a novel experimental apparatus to measure accurately the rheological response of non-Newtonian fluids under shear-free conditions that are characteristic of those experienced in the containerless processing of materials. The instrumentation generates a homogeneous, uniaxial fluid filament through rapid, exponential stretching of a liquid bridge. The spatial uniformity of the deformation experienced by the fluid sample is verified by digital particle image velocimetry (DPIV) and videoimaging. Direct measurements of the tensile force exerted by the material and the column diameter then allow calculation of the transient extensional stress growth in the fluid as a function of strain and strain-rate. In ground-based tests, the evolution in the internal microstructure of the polymer chains is also followed using a phase-modulated laser polarimeter to measure the flow-induced birefringence of the polymer solution. The		

Task Description:	apparatus is a flexible and adaptable scientific instrument that can accommodate various classes of non-Newtonian liquids and will also be suitable for multiple flight capabilities or long term multi-user operation aboard a permanent orbital platform. The objectives of this work are to (i) successfully generate a simple homogeneous shear free flow in the material while measuring the kinematic and dynamical response of the material during this constant strain rate deformation; (ii) determine the transient extensional viscosity in a uniaxial stretching flow for dilute polymer solutions and the subsequent relaxation behavior after cessation of extensional deformation; and (iii) (Ground instrumentation only) to measure the evolution of the molecular orientation of the polymer chains using flow-induced birefringence during the deformation and relaxation. The extensional viscosity is a fundamental physical property of all non-Newtonian fluid materials, and cannot be determined from simple viscometric shear flow experiments. Constitutive equations for viscoelastic fluids such as dilute polymer solutions predict large changes in the extensional viscosity as the elongation rate is increased; however, the validity of these theories is complicated by the paucity of experimental data obtained in extensional flows. Most quantitative rheologicalmeasurements in shear-free flows have only been performed using highly elastic or "stiff" materials such as polymer melts which can easily be elongated without sagging under a gravitational body force. By performing similar experiments in an extended microgravity environment it is possible for the first time to obtain accurate measurements of the extensional viscosity for more 'mobile' fluids such as polymer solutions, suspensions and liquid crystalline materials. This rheological data will allow designers of both space- and ground-based material processes to use improved constitutive models in numerical simulations of complex two- and three-dimensional fluid flows. In a simple stretching flow, fluids such as water or syrup exhibit a resistance to stretching that is exactly three times the value of their Newtonian viscosity. This resistance is usually termed the "extensional viscosity" of the material. However, for fluids containing long macromolecules (e.g., synthetic polymers, liquid crystals, or DNA) it is predicted that the extensional viscosity can be anywhere from 100 to 10000 times greater than the normal shear viscosity of the fluid. In the near future, NASA plans to develop "containerless processing" operations under microgravity and a detailed understanding of extensional properties in fluids will be absolutely critical -- since, in the absence of container walls, the only way to mix, pour, and shape fluids will be through pure stretching motions. Knowledge of such nonlinear material functions is fundamental to the development and verification of relatively cheap ground-based computational modeling techniques which can be used effectively in the "a priori" design of advanced microgravity material processes - thereby alleviating the need for costly in-flight pilot experiments. Finally, in addition to their significance in many industrial processes such as fiber-spinning, spraying and film coating operations, stretching flows of non-Newtonian fluids are of fundamental scientific importance to a number of complex fluid dynamics phenomena, including the stability and breakup of jets, enhanced oil recovery and turbulent drag reduction for advanced aircraft, boats, and submarines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In the final year of the contract, we modified the Filament Stretching Elongational Rheometer (FISER) instrumentation to enable the testing of adhesive like materials. Typically Pressure Sensitive Adhesives (PSAs) are tested in ‘tack-testers’ under conditions of narrow gaps and linear stretching profiles. It is difficult in such geometries to extract quantitative information about the material properties. As part of a Master’s thesis, Mr. Ryan Welsh developed hardware fixtures that enabled the FISER to probe such materials using a well-controlled exponential profile. A prism-configuration furthermore enabled the endplate instability observed by Spiegelberg et al. (1996) to be investigated in detail. This MS Thesis has been widely distributed amongst the pressure-sensitive adhesive community, and a long publication is presently being completed. Finally it should be noted that the innovative work performed by the students and postdoctoral researcher supported by this grant has been recognized by the research community. Three students/postdocs funded by this NASA research have now accepted junior faculty positions at tier-1 US Universities. One paper published in 2001 under this NASA grant by J. Rothstein and the P.I. was awarded the Frenkiel Award of the American Physical Society – Division of Fluid Dynamics in the Fall of 2002.
Bibliography Type:	Description: (Last Updated: 09/04/2025)